



DELHI PUBLIC SCHOOL NEWTOWN

SESSION 2021-22

MONDAY TEST

CLASS: IX
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 28/ 05/ 2021

GENERAL INSTRUCTIONS:

- The paper consists of one printed page.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

1. Find the amount and the compound interest on ₹ 1,00,000 for $1\frac{1}{2}$ years at 8% per annum, the interest being compounded semi-annually. [4]
2. If $p + q + r = 12$ and $p^2 + q^2 + r^2 = 50$, find $pq + qr + pr$. [4]
3. Convert $4.\overline{6274}$ into vulgar fraction. [4]
4. If $a + 3b + 2c = 0$, Prove that $a^3 + 27b^3 + 8c^3 = 18abc$ [4]
5. Prove that $\sqrt{17}$ is an irrational number by the method of contradiction. [4]
6. If $x = \frac{1}{4-x}$, find the value of (i) $x + \frac{1}{x}$ (ii) $x^2 - \frac{1}{x^2}$ (iii) $x^3 + \frac{1}{x^3}$ [5]
7. A man invests ₹ 10,000 for three years at a certain rate of interest, compounded annually. At the end of one year it amounts to ₹ 10,600. Calculate:
(i) the rate of interest per annum
(ii) the interest accrued in the second year.
(iii) the amount at the end of the third year. [5]
8. The simple interest on a certain sum for 4 years is ₹ 2400 and the compound interest on the same sum at the same rate for 2 years is ₹ 1230. Find
(i) the rate of interest,
(ii) the principal. [5]
9. Simplify: $\frac{4\sqrt{3}}{2-\sqrt{2}} - \frac{30}{4\sqrt{3}-3\sqrt{2}} - \frac{3\sqrt{2}}{3+2\sqrt{3}}$ [5]